

1.04 Sequences and Series				
1.04a 1.04c	Binomial expansion	a) Understand and be able to use the binomial expansion of $(a + bx)^n$ for positive integer n and the notations $n!$ and nC_r, ${}_nC_r$ or $\binom{n}{r}$, with ${}^nC_0 = {}^nC_n = 1$. <i>e.g. Find the coefficient of the x^3 term in the expansion of $(2 - 3x)^7$.</i> <i>Learners should be able to calculate binomial coefficients. They should also know the relationship of the binomial coefficients to Pascal's triangle and their use in a binomial expansion.</i>	c) Be able to extend the binomial expansion of $(a + bx)^n$ to any rational n, including its use for approximation. <i>Learners may be asked to find a particular term, but the general term will not be required.</i> <i>Learners should be able to write $(a + bx)^n$ in the form $a^n \left(1 + \frac{bx}{a}\right)^n$ prior to expansion.</i>	MD1
1.04b 1.04d		<i>They should also know that $0! = 1$.</i> b) Understand and know the link to binomial probabilities.	d) Know that the expansion is valid for $\left \frac{bx}{a}\right < 1$. <i>[The proof is not required.]</i> <i>e.g. Find the coefficient of the x^3 term in the expansion of $(2 - 3x)^{\frac{1}{3}}$ and state the range of values for which the expansion is valid.</i>	

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1.04e	Sequences		e) Be able to work with sequences including those given by a formula for the nth term and those generated by a simple relation of the form $x_{n+1} = f(x_n)$. <i>Learners may be asked to generate terms, find nth terms and comment on the mathematical behaviour of the sequence.</i>	MD2
1.04f			f) Understand the meaning of and work with increasing sequences, decreasing sequences and periodic sequences. <i>Learners should know the difference between and be able to recognise:</i> <i>1. a sequence and a series,</i> <i>2. finite and infinite sequences.</i>	
1.04g	Sigma notation		g) Understand and be able to use sigma notation for sums of series.	MD3
1.04h	Arithmetic sequences		h) Understand and be able to work with arithmetic sequences and series, including the formulae for the nth term and the sum to n terms. <i>The term arithmetic progression (AP) may also be used. The first term will usually be denoted by a, the last term by l and the common difference by d. The sum to n terms will usually be denoted by S_n.</i>	MD4

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1.04i 1.04j	Geometric sequences		i) Understand and be able to work with geometric sequences and series including the formulae for the nth term and the sum of a finite geometric series. <i>Learners should know the difference between convergent and divergent geometric sequences and series.</i> j) Understand and be able to work with the sum to infinity of a convergent geometric series, including the use of $r < 1$ and the use of modulus notation in the condition for convergence. <i>The term geometric progression (GP) may also be used. The first term will usually be denoted by a and the common ratio by r. The sum to n terms will usually be denoted by S_n and the sum to infinity by S_∞.</i>	MD5
1.04k	Modelling		k) Be able to use sequences and series in modelling. <i>e.g. Contexts involving compound and simple interest on bank deposits, loans, mortgages, etc. and other contexts in which growth or decay can be modelled by an arithmetic or geometric sequence.</i> <i>Includes solving inequalities involving exponentials and logarithms.</i>	MD6

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